

Group recovery after trimming, and level-free flagging, in robust clusterwise regression

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Abstract

Trimming methods for robust clusterwise regression discard a fixed fraction of the data. Too low a level breaks the fit; too generous a level can trim away a small group. We first propose a group-recovery step that can follow any trimming or flagging method: it searches the discarded units for a line, tests whether those near it form a peak rather than a band, and restores the line as a group when the likelihood of Gaussian groups plus uniform noise improves by a margin like that of the Bayesian information criterion. In simulations it repaired the failures of a generous TCLUS-REG level with unequal groups, but not with three or four groups. The second proposal, ESF (exact-subsample flagging), is a flagging procedure without a trimming level: it solves the clusterwise least-squares problem exactly on small subsamples, flags units far from the best fit, and draws later subsamples from the rest. Two constants stand in for the level: a subsample size, set from a lower bound on the smallest group proportion, and a cap on the flagged set. It is meant for data about whose contamination nothing is known: TCLUS-REG at the fixed level 0.30, followed by reweighting and the recovery step, was as accurate as ESF on average up to a fifth of outliers, and higher levels were more accurate beyond. The flagged fraction estimates the contamination only when the errors are close to Gaussian. On taxi fares with known tariffs, ESF found both in every sample.

Keywords: clusterwise regression, robust clustering, trimming, reweighting, outliers, branch and bound

1. Introduction

In clusterwise linear regression each observation follows one of K linear models, and the model that generated it is not observed. For units (x_i, y_i) , $i = 1, \dots, n$, with $x_i \in \mathbb{R}^d$ including an intercept,

$$y_i = x_i^\top \theta_{z_i}^0 + \sigma_{z_i} e_i, \quad z_i \in \{1, \dots, K\}, \quad (1)$$

where z_i is the unobserved label of unit i , $\theta_k^0 \in \mathbb{R}^d$ and $\sigma_k > 0$ are the coefficients and noise scale of group k , and the e_i are independent standard errors. The model goes back to Quandt

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(1972) and Späth (1979). It is usually fitted by least squares over partitions or by maximum likelihood for a mixture of regressions (DeSarbo & Cron, 1988; De Veaux, 1989).

Both fits break down when a fraction of the data does not follow any of the K models: a few gross outliers in the response can pull a fitted line far from every group, or be fitted as a group of their own. The standard remedy is trimming. TCLUST-REG (García-Escudero et al., 2010) discards the fraction α of units that fit worst and fits the model to the rest; the trimmed likelihood estimator (TLE) of Neykov et al. (2007) and the trimmed cluster-weighted model (García-Escudero et al., 2017) do the same within other likelihoods. Each needs the trimming level α , which the analyst rarely knows. If α is smaller than the fraction of outliers, the fit breaks down; if it is much larger than that fraction, genuine units are discarded, and a small group can be trimmed away altogether.

Several ways around this choice have been proposed. One trims at a deliberately high level and then reinstates the units that are well fitted, a reweighting step proposed for robust clustering by Dotto et al. (2018); it needs a level that is high enough. Another monitors the fit over a grid of levels and leaves the choice to the analyst (Torti et al., 2019; Riani et al., 2008), or scans the number of groups and the level jointly by trimmed likelihood curves (García-Escudero et al., 2011; Cerioli et al., 2018). A third adapts the trimming within each group (Chang et al., 2020), and a fourth replaces trimming by bounded loss functions (Bai et al., 2012). Mixtures with a mean-shift parameter for each unit (Yu et al., 2017), with a uniform noise component (Banfield & Raftery, 1993), or with contaminated normal errors (Punzo & McNicholas, 2017; Mazza & Punzo, 2020) model the outliers instead of trimming them. Multiple-model fitting from random minimal samples, from RANSAC (Fischler & Bolles, 1981) to J-linkage (Toldo & Fusiello, 2008), T-linkage (Magri & Fusiello, 2014) and preference analysis (Magri & Fusiello, 2015), treats a related problem with the noise scale as a user-set threshold, and does not ask whether what it discards is a missed model.

This paper makes two proposals. The first concerns what a robust fit throws away. When one group is small, a trimming method at a generous level can trim it away with the outliers, and a fit that flags units by their residuals can spend two lines on the large group and flag the small one; our simulations show both happening to a group of 20% of the data even without outliers. The *group-recovery step* (Algorithm 2) searches the trimmed or flagged units for a line around which many of them are concentrated, and keeps the line if the likelihood of Gaussian groups plus uniform noise prefers the fit that contains it. The step needs only a fit and its flagged set, so it can follow TCLUST-REG with reweighting as well as the level-free procedure introduced next; it is what makes a generous level safe with unequal groups. Trimmed likelihood curves ask the same question by a joint scan; the step asks it locally and post hoc (Section 2.4).

The second proposal is a flagging procedure that needs no level, the exact-subsample flagging procedure, ESF (Algorithm 1). A small random subsample that contains no outlier, fitted by least squares over all its partitions, gives a clean fit of the model, and that fit identifies the outliers of the whole sample as the units far from every fitted line. ESF therefore draws many small subsamples, fits each one exactly, scores each fit by a bounded loss on the units not yet flagged, and flags the units that lie far from the best fit; later subsamples are drawn only from units not yet flagged. The fit ends with the reweighting step of robust regression (Rousseeuw & Leroy, 1987), and outlying covariates are flagged by a separate screen. A unit is flagged when its residual exceeds a fixed multiple of a robust

scale; this cut-off takes the place of the trimming level, and it does not depend on how many outliers there are. Two constants stand in for the level: the subsample size m , set from a lower bound on the smallest group proportion, and the cap of $n/3$ on the flagged set beyond which the recovery step is not attempted; Section 2.5 gives a default rule for each. Subsampling to find a clean subset is the idea behind least median of squares and least trimmed squares (Rousseeuw, 1984; Rousseeuw & Van Driessen, 2006), behind robust linear grouping (García-Escudero et al., 2009) and behind the random starts of TCLUST-REG. Here each subsample contains several units of every group and is solved to global optimality, so a clean subsample yields the clusterwise least-squares fit of that subsample and not a fit to d points. The exact fit is the problem that the operations research literature solves on whole samples (Bertsimas & Shioda, 2007; Carbonneau et al., 2011, 2012, 2014; Fois et al., 2026; Bagirov et al., 2013; Joki et al., 2020; Park et al., 2017), without treating outliers. Here it is applied about a hundred times per data set to subsamples of a dozen units, where a plain depth-first branch and bound is one to two orders of magnitude faster than a mixed-integer programme. The accuracy of ESF does not rest on the exactness: with ten starts of hard alternation in place of the exact fit of each subsample, ESF alone was within 0.01 of the exact fit in most cells, ahead by 0.025 in one cell and behind by 0.02 and 0.04 with four groups (Supplement, Table S14). The exact fit removes instead the dependence of a subsample’s fit on its starting point.

The place of ESF among the trimming methods should be stated at once. In the simulation study, TCLUST-REG at the fixed level 0.30 with reweighting and the recovery step matched the mean accuracy of ESF with up to 20% of outliers and had no cell below 0.85, where ESF had two; the level 0.25 with equal group weights did better still; and the levels 0.35 and 0.40 beat ESF beyond 25% of outliers (Table 2). The level 0.35 came closest to serving both ranges; up to 20% of outliers it fell behind ESF only with three and four groups, and at 35% it fell behind in four designs with two groups. ESF is therefore not the more accurate method; its case is that it needs no level and holds up with three and four groups, where every fixed level with estimated group weights loses accuracy.

The contributions are the following.

- (i) The group-recovery step, which can follow any trimming or flagging method (Section 2.4).
- (ii) ESF (Section 2), with a comparison of exact solvers for the small clusterwise least-squares problems and an ablation of the exact fit (Section 2.2).
- (iii) Three partial results (Section 3): the exact probability that a stage of the sequential draw contains a clean subsample, the large-sample limit of the final reweighting step and of the fraction it flags, and the value of the two operations of the recovery step in the population.
- (iv) A simulation study on 7,750 data sets (Section 4). After TCLUST-REG at a generous level with reweighting, the recovery step repaired nearly all failures with unequal or small groups but not those with three or four groups. ESF followed by the recovery step was within 0.03 of the better of the levels 0.25 and 0.40, chosen after the run, in 83 of 89 cells with up to a fifth of outliers, and was the less robust choice beyond a quarter.

- (v) Three data sets (Section 5): the fishery and tone perception data, and taxi fares from an airport, where the tariff of every trip is recorded and so the true group of every unit is known.

No method in the study is best everywhere; Section 6 says which to use when. The constants of ESF and of the recovery step were the same in every experiment.

2. ESF and the recovery step

We write ESF for the flagging procedure of Algorithm 1. Unless stated otherwise, ESF is followed by the recovery step of Algorithm 2, and *ESF alone* denotes Algorithm 1 without it. The recovery step (Section 2.4) is described after the fit whose flagged set it examines.

2.1. Exact fits on small subsamples

For a set S of units, the least-squares clusterwise fit of S solves

$$\min_{z \in \{1, \dots, K\}^S} \min_{\theta \in \mathbb{R}^{Kd}} \sum_{i \in S} (y_i - x_i^\top \theta_{z_i})^2. \quad (2)$$

For fixed labels the inner problem separates into K ordinary least-squares fits, so the difficulty lies in the labels alone. On the whole sample the problem is hard: whether n points in the plane can be covered by K lines, that is whether (2) with one covariate has value zero, is NP-complete when K is part of the input (Megiddo & Tamir, 1982). On a subsample of a few dozen units it can be solved exactly by a depth-first branch and bound over label vectors, which prunes a partial labelling as soon as the sum of the residual sums of squares of its groups exceeds the best value found (Supplement, Section S7). The cost does not depend on n , but it grows quickly with K and m (Supplement, Table S30), which limits ESF to a small number of groups.

For fixed coefficients the best label of each unit is its nearest line, so the exact fit of a subsample is a K -means fit with lines in place of centres (Pollard, 1981); here and below a *line* is the graph of $x \mapsto x^\top \theta_k$, a hyperplane when there is more than one covariate. The same rule, $h_\theta(u) = \arg \min_k (y - x^\top \theta_k)^2$ for a unit $u = (x, y)$, with ties broken by the smallest index, extends the fit to the whole sample. We call the fit of one subsample a *replicate*; it is *admissible* if each of its K groups contains at least $d + 1$ units, and inadmissible replicates are discarded and redrawn, which forces $m\pi_{\min} \gtrsim d + 1$ for the subsample size m and the smallest group proportion $\pi_{\min}$.

2.2. Cost of the exact fit

ESF needs about a hundred solves of (2) on a dozen units per data set, so what matters is the cost of one small solve. On the same subsamples we compared our depth-first search, which also prunes partial labellings that cannot give every group $d + 1$ units, with repetitive branch and bound (Brusco, 2006) and with the mixed logical-quadratic programme of Carboneau et al. (2011) solved by Gurobi 13.0 (Gurobi Optimization, 2026) and by SCIP 10.0 (Maher et al., 2016) (Supplement, Section S4 and Tables S26 and S30). For $m \leq 20$ the depth-first search was 30 to 180 times faster than Gurobi, and further still than SCIP, which solves a different, big- M formulation of the programme. The gap closes as m grows, and the

Algorithm 1 ESF: clusterwise regression from exactly solved subsamples, with sequential flagging

Require: data $(x_i, y_i)_{i \leq n}$; number of groups K ; subsample size m ; replicates B ; stages L ; weight λ ; cut-offs c, c_b

- 1: $\mathcal{F} \leftarrow$ units whose covariates are outlying (robust distance above the 0.975 quantile of χ_p^2 , p the number of covariates)
- 2: **for** $\ell = 1, \dots, L$ **do**
- 3: draw B/L admissible replicates, on subsamples of size m drawn uniformly from $\{1, \dots, n\} \setminus \mathcal{F}$, each solved exactly
- 4: $r_{bi} \leftarrow \min_k (y_i - x_i^\top \hat{\theta}_k^{(b)})^2$, the squared residual of unit i from its nearest line of replicate b , for every replicate b so far and every unit i
- 5: $s^2 \leftarrow \min_b \text{med}_{i \notin \mathcal{F}} r_{bi} / 0.4549$, the median taken over the unflagged units > 0.4549 is the median of a χ_1^2 variable
- 6: $F_b \leftarrow \sum_{i \notin \mathcal{F}} \min\{r_{bi}, c_b^2 s^2\}$; $w_b \leftarrow \exp\{-\lambda(F_b - \min_{b'} F_{b'}) / [(n - |\mathcal{F}|) s^2]\}$
- 7: reference $\leftarrow$ replicate $b^* = \arg \min_b F_b$
- 8: $\mathcal{F} \leftarrow \{i : i \text{ is far from the reference in the response or in the covariates}\}$ (*Flagging below*)
- 9: **end for**
- 10: labels $\tilde{z} \leftarrow$ plurality of the nearest-line labels of all replicates, weighted by w
- 11: refit each group on its unflagged units; concentration steps keeping $n - |\mathcal{F}|$ units
- 12: reweighting: recompute the flags at the current fit and refit on the unflagged units
- 13: **return** coefficients $\hat{\theta}$, labels $h_{\hat{\theta}}(u_i)$ for every unit, the flags $\mathcal{F}$ recomputed at $\hat{\theta}$, and $\hat{\alpha} = |\mathcal{F}|/n$

programme overtakes the search from about $m = 40$ with two groups; the repetitive bound made the search slower at every size tried. Exactness matters for accuracy less than for reliability (Section 1; Supplement, Table S14): with one start of hard alternation in place of the exact fit, ESF alone lost up to 0.11 with four groups, and with elemental fits as in RANSAC up to 0.25.

2.3. Sequential flagging

Algorithm 1 states ESF. It runs in L stages, each drawing B/L admissible replicates from the units not yet flagged; after each stage all replicates so far are scored by a bounded loss on those units, and the units far from the best-scoring replicate are flagged. The flags are recomputed at every stage, so a unit flagged at one stage can be reinstated at the next.

Flagging. A unit is far from a fit θ in the response if its absolute residual from its nearest line exceeds c times the scale of the group of that line. That scale is 1.4826 times the median absolute residual of the units assigned to the line (or of all units, when fewer than ten are); with a scale per group, a group with small noise does not hide outliers and a group with large noise is not trimmed. A unit is far in the covariates if its robust distance from the other units of its group, computed with the minimum covariance determinant (Rousseeuw & Van Driessen, 1999), exceeds the 0.975 quantile of χ_p^2 ; the test is applied to groups with at least $\max(10, 5p)$ units, and at the start, before any fit, to all units as one group. It is the

covariate trimming of TCLUST-REG (García-Escudero et al., 2010) with a fixed quantile in place of a trimming level.

Scoring. The bounded loss caps each squared residual at $c_b^2 s^2$, where s is a scale common to all replicates, so a replicate that fits the clean units well has a low score whatever the outliers do. The scale is the smallest, over the replicates, of the median squared residual over all unflagged units, scaled to estimate the error variance of a replicate that fits the clean majority (Supplement, Section S7). The criterion has a weakness that more search does not cure. With a small group, the bounded loss with the best replicate’s scale can prefer two nearly parallel lines on the large group, whose units they fit more tightly, to one line on each group, and a wider search finds such fits more often: with $B = 500$ replicates in place of 100, the cell means of the small-group designs at 30% of outliers fell by 0.04 to 0.07 (Supplement, Table S29), and the recovery step then has to undo what the scoring chose. A less biased scale did not help (Supplement, Section S6.2).

Labels and final fit. The labels of the final fit are the weighted plurality of the labels that the replicates give to each unit, after alignment with a pilot replicate that serves only for alignment (Supplement, Section S7). The concentration steps are those of least trimmed squares (Rousseeuw & Van Driessen, 2006) and TCLUST-REG, with the number of retained units set by the flags rather than by a level, and the last step is the reweighting step of robust regression (Rousseeuw & Leroy, 1987; Gervini & Yohai, 2002); every unit then receives the label of its nearest fitted line.

2.4. Is the flagged set a group?

ESF flags every unit that its K lines do not explain, and it cannot tell a unit that belongs to no group from a unit of a group it has missed. The second case arises when one group is small: the best-scoring fit may spend two nearly parallel lines on the large group and flag most of the small group (Section 4.5; Figure 1). A trimming method at a generous level does the same by construction. The recovery step, Algorithm 2, examines the flagged set for this case. It uses only a fit $\hat{\theta}$, its flagged set $\mathcal{F}$ and the group scales s_k of the flagging rule, so it applies after ESF and equally after TCLUST-REG with reweighting. In the algorithm, $\mathcal{F}'$ is $\mathcal{F}$ without the units of the initial covariate screen, which cannot form a line in the response. The weight w is that of the Gaussian component in a two-component fit to the residuals of the flagged units near the candidate line; this fit is the *peak test*.

The steps have simple reasons. A missing group lies close to one line, so a line is sought among the flagged units, as in RANSAC (Fischler & Bolles, 1981). Gross outliers also contain lines that pass near many of them, so most flagged units near a candidate must sit in a peak of the width of the fitted groups, not in a band. The comparison uses the likelihood of the model that the flagging rule assumes, Gaussian groups and flagged units spread over the range of the response: two lines through one group describe it hardly better than one line, whereas a group of n_1 units described as noise instead of by its own line loses about $n_1\{\log(\pi_1 R/\pi_0) - \log \sigma_1 - 1.42\}$ in log-likelihood (Proposition 3). The noise range R is set by the extreme responses, so gross outliers raise the value of a group; a range taken over the unflagged units alone lost small groups (Supplement, Section S6). The margin is a heuristic modelled on the Bayesian information criterion. The restriction on the scales of a candidate fit is that of TCLUST-REG: the factor 12 bounds the ratio of the group variances, that is $\sqrt{12}$ on the scales, as in the TCLUST-REG runs of Section 4, and it prevents a candidate

Algorithm 2 The group-recovery step, applied to a fit and its flagged set

- Require:** fit $\hat{\theta}$ with flags $\mathcal{F}$ and group scales $s_1, \dots, s_K$; noise range $R = 1.1 (\max_i y_i - \min_i y_i)$; margin $\frac{d+1}{2} \log n$; restriction factor 12 on the variances
- 1: stop if $|\mathcal{F}| > n/3$; otherwise let $\mathcal{F}'$ be $\mathcal{F}$ without the units of the initial covariate screen, and $s = \text{med}_k s_k$
 - 2: fit a line β to $\mathcal{F}'$: the line through d units of $\mathcal{F}'$, among 500 random choices, that has most units of $\mathcal{F}'$ within cs , refitted three times by least squares on the units within cs
 - 3: stop unless at least $\max\{2(d+1), 0.03n\}$ units of $\mathcal{F}'$ lie within cs of β and they form a peak: in the model $w\mathcal{N}(0, s^2) + (1-w)\text{U}(-3cs, 3cs)$ for the residuals from β of the units of $\mathcal{F}'$ within $3cs$, the fitted w is at least $\frac{1}{2}$
 - 4: **for** $j = 1, \dots, K$ **do**
 - 5: $\theta^{(j)} \leftarrow$ the lines of $\hat{\theta}$ without line j , together with β , refitted by the reweighting step
 - 6: **end for**
 - 7: score every fit by the log-likelihood ℓ of K Gaussian lines plus a uniform noise component of density π_0/R , with the scales, proportions and noise weight π_0 read off the fit (the units of the initial covariate screen count as noise)
 - 8: among the $\theta^{(j)}$ whose scales are all at most $\sqrt{12} \min_k s_k$, with s_k the scales of $\hat{\theta}$, take the one with the largest ℓ ; it replaces $\hat{\theta}$ if $\ell(\theta^{(j)}) - \ell(\hat{\theta})$ exceeds the margin
 - 9: apply the step again while the fit changes, at most three times in all
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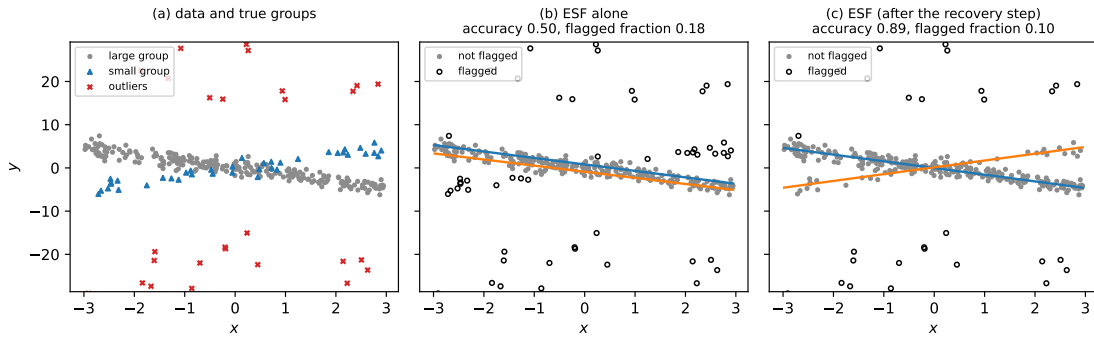


Figure 1: The recovery step on one simulated data set of the third part of the study (groups of 85% and 15% of the clean units, $n = 300$, $\varepsilon = 0.10$, 28 outliers). (a) The true groups. (b) The fit of ESF alone: two nearly parallel lines on the large group, with most of the small group flagged. (c) After the recovery step. Flagged units lie more than 2.5 robust scales from every line.

from absorbing the contamination into a wide group. The peak test does not exclude every band: a band of uniform noise of half-width below about 3.4 scales passes it (Supplement, Section S2.3), and after a generous trimming level the step accepted such bands at high contamination (Section 4.6). The step is not attempted when a third of the units or more are flagged, because the fit it would examine is then itself in doubt. The constants are fixed, not fitted to the data (Section 2.5).

Relation to trimmed likelihood curves. The question the step asks, whether the discarded units hide a group, is the one that classification trimmed likelihood curves (García-Escudero et al., 2011) and the constrained-likelihood approach of Cerioli et al. (2018) answer for model-based clustering by fitting the model over a grid of pairs (K, α) and reading the number of groups and the level off the curves. For clusterwise regression FSDA’s `tclustregIC` (Riani et al., 2012; Torti et al., 2021) chooses the number of groups and the restriction factor by an information criterion at a given level, and the level itself is monitored over a grid (Torti et al., 2019); the level is thus still the analyst’s choice. The recovery step answers that question locally, post hoc and for a single candidate: it needs no grid over α , and it works after a method that has no α , such as ESF or a reweighted fit; but it tests one candidate at a time, at the K of the fit, and cannot tell that the fit has one group too many or that the level was too low. When the number of groups is in doubt, a scan over K such as that of `tclustregIC` is the better tool, and the step is then a check on what the chosen level trimmed.

2.5. Tuning constants and defaults

Supplement Table S44 lists the constants of both algorithms, the values used in every experiment, the range examined and the largest change in a cell’s mean accuracy over that range. Up to 20% of outliers, moving one constant over its range changed a cell mean by at most 0.05, with four exceptions: the loss cap (0.13 lost at $c_b = 5$ in D5, while ESF alone gains 0.24 with a group of 15% and no outliers, a gain we obtain instead through the recovery step), the covariate quantile with high-leverage points (0.21), the limit on the flagged fraction (0.20 at $0.25n$) and $m = 12$ in every design (0.07 lost with four groups); $B = 500$ replicates moved cell means by -0.02 to $+0.03$ up to 20% and by up to -0.07 at 30%.

The two constants that stand in for the level. The subsample size must allow every group $d + 1$ units, so $m \geq (d + 1)/\pi_{\min}$; larger values make clean replicates more accurate but rarer, since a subsample is clean with probability close to $(1 - \varepsilon)^m$. We used $m = 8$ for two groups and one covariate, $m = 10$ when one group has 30% of the units, and $m = 12$ for three groups or three covariates. The default rule is to take m near $(d + 1)/\pi_{\min}$, rounded to an even number, for the lower bound $\pi_{\min}$ on the smallest group proportion that the analyst is prepared to assume, and $m = 12$ when no bound is available. A lower bound on $\pi_{\min}$ is prior knowledge of the same kind as an upper bound on ε , and the competitors are not given it, so ESF with $m = 12$ in every design was run on all data sets and appears as a row of Tables 1, 2 and 3. Over the 89 cells up to 20% of outliers its mean accuracy is 0.912 against 0.914, with the same two cells below 0.85, and beyond 25% it is 0.813 against 0.820. The loss is concentrated in the four-group design (0.88 against 0.95), where $m = 12$ leaves three units per group, and in clustered outliers at $\varepsilon = 0.20$ (0.75 against 0.78). With small groups the recovery step makes up for the smaller subsamples, which cost ESF alone more (mean

0.864 against 0.874 over the 89 cells up to 20%; Supplement, Tables S12, S15, S28 and S39). The cap of $n/3$ on the flagged set is a guard against applying the recovery step to a fit that has broken down, not a trimming level: moving it to $0.25n$ lost 0.16 and 0.20 in the two small-group designs at $\varepsilon = 0.20$, where ESF alone flags about a quarter of the units, and moving it to $0.40n$ gained up to 0.05 in those designs at $\varepsilon = 0.30$ and moved no other cell by more than 0.01 (Supplement, Table S23 and Section S5). A higher cap is reasonable only when the contamination is known to be below a third and a small group is suspected.

Other constants. The vote weight is $\lambda = 3$, with $B = 100$ replicates in $L = 10$ stages (Section 3). The cut-off $c = 2.5$ is the usual value for reweighting in robust regression (Rousseeuw & Leroy, 1987): at the true parameter a clean unit is then flagged with probability $2\Phi(-2.5) \approx 0.012$ (Proposition 2). The cap $c_b = 3$ lies slightly above c , so that a unit at the flagging boundary still contributes its full residual.

A default for real data. The study used $B = 100$ replicates and one seed per data set, and on the fishery data (Section 5.1) the answer depended on the seed. The default for real data is $B = 100$ with the best of five seeds by the likelihood of the model of the recovery step, a rule that returned the modal fit every time on the fishery data and, on the simulated data, raised cell means up to 20% of outliers and lowered one cell beyond (Supplement, Section S6.3). When a group is fitted exactly, as the flat tariff of Section 5.3, the likelihood rewards any line through tied points and the seeds should be compared by eye. The flagged fraction $\hat{\alpha}$ is the check on the result. A value well below any plausible contamination means that the outliers have been absorbed into a fitted line and the fit has broken down (Section 4.3). A value well above it, as on the tone and taxi data, means that the errors are not close to Gaussian or that a group's covariates are compact; $\hat{\alpha}$ is then not an estimate of the contamination.

3. Theoretical results

Three partial results follow; their precise statements, proofs and scope are in Section S2 of the Supplement, and none covers the passage from one stage of ESF to the next, the weighted vote, or the search for the candidate line. Throughout, $O \subseteq \{1, \dots, n\}$ is a fixed set of contaminated units whose values are arbitrary, and a subsample is *clean* if it contains no unit of O .

One stage of the sequential draw. Let E_ℓ be the set of units not flagged after stage $\ell - 1$, from which stage ℓ draws, and $\varepsilon_\ell = |E_\ell \cap O|/|E_\ell|$.

Proposition 1 (informal statement; the precise version is in Supplement S2.0). *Conditionally on the data and on everything drawn before stage ℓ , the subsamples of stage ℓ are independent and uniform on the admissible subsamples of size m of E_ℓ , each replicate is clean with probability $\pi_\ell = p_\ell a_\ell$, where $p_\ell \leq (1 - \varepsilon_\ell)^m$ is the probability that a uniform subsample of E_ℓ is clean and a_ℓ corrects for admissibility, and the stage contains a clean replicate with probability $1 - (1 - \pi_\ell)^{B/L}$.*

The result is exact and covers one stage: a clean replicate becomes likelier as the flags remove outliers from E_ℓ , but nothing here says that earlier stages do so. With $m = 8$ the first stage contains a clean replicate with probability about 0.84 at $\varepsilon = 0.20$ and 0.45 at 0.30, and with $m = 12$ about 0.51 and 0.13, which is where ESF breaks down.

The final reweighting step. The reweighting step takes a parameter θ , recomputes the flags and refits each group on its unflagged units. Let the units be independent with law $P = (1 - \varepsilon)P_0 + \varepsilon C$, where under P_0 a unit of group k , drawn with probability $\pi_k > 0$, has $y = x^\top \theta_k^0 + \sigma_k e$ with $e \sim \mathcal{N}(0, 1)$ independent of x , and every unit of C lies beyond $c' > c$ scales from the nearest line; let η be the probability that a clean unit is nearer to another true line than to its own.

Proposition 2 (informal statement; the precise version is Theorem S2 of the Supplement). *Apply the map at $\theta = \theta^0$. Under mild regularity conditions, almost surely as $n \rightarrow \infty$, the scale of group k converges to a limit s_k with $s_k/\sigma_k = 1 + O(\nu_k)$, ν_k being of the order of $\eta + \varepsilon$ divided by the mass assigned to line k ; the flagged fraction converges to $\varepsilon + (1 - \varepsilon)\kappa$, where κ , the fraction of clean units flagged, lies within η of $\sum_k \pi_k 2\Phi(-cs_k/\sigma_k)$; and the refitted coefficients converge to a limit that depends on the contamination only through the s_k and lies within $O(\eta)$ of θ_k^0 when the covariates are bounded. Without overlap or contamination, θ^0 is a fixed point of the map and the flagged fraction tends to $2\Phi(-c)$.*

This is a law of large numbers for one application of the map at a fixed input: it says what the flagged fraction estimates under Gaussian errors, ε plus a fraction of clean units close to $2\Phi(-c) \approx 0.012$ when the groups overlap little, and that contamination beyond the cut-off is removed entirely. It does not cover four things: a small group under moderate contamination, where the condition $\nu_k < \frac{1}{2}$ of Theorem S2 fails; the input that ESF delivers to the map; the repeated application of the map in the code; and the flagged fraction reported at the refitted parameter (Supplement, Section S2.2).

The group-recovery step. The recovery step scores a fit F , a set of lines θ_k with scales s_k and proportions π_k and a noise weight $\pi_0 > 0$ summing to one, by $\ell_n(F) = \sum_{i \leq n} \log f_F(u_i)$, where $f_F(u) = \sum_k (\pi_k/s_k) \varphi\{(y - x^\top \theta_k)/s_k\} + \pi_0/R$. For given parameters, $\ell_n(F)/n$ converges almost surely to $L(F) = \mathbb{E}_P \log f_F(u)$; the proposition values, in L , the two operations by which a candidate differs from the original fit.

Proposition 3 (informal statement; the precise version is in Supplement S2.3). *(i) Let G be an event of probability π_* on which $y = x^\top \theta_* + \sigma_* e$ with $e \sim \mathcal{N}(0, 1)$ independent of x , let F have noise weight $\pi_0 > \pi_*$, and let F' be F with the line $(\theta_*, \sigma_*, \pi_*)$ added and the noise weight lowered to $\pi_0 - \pi_*$. Then*

$$L(F') - L(F) \geq \pi_* \left\{ \log \frac{\pi_* R}{\pi_0} - \log \sigma_* - \frac{1}{2} \log(2\pi e) - \xi \right\} - \nu \log \frac{\pi_0}{\pi_0 - \pi_*}, \quad (3)$$

where $\xi \geq 0$ is small when G lies a few scales from every line of F , and $\nu \leq 1 - \pi_$ is the noise mass that F attributes to the units outside G . (ii) If F contains two lines through one group of probability π_S and scale σ_S , at intercepts δ apart, and F'' replaces them by the line through the group, then $|L(F'') - L(F)| \leq (\varphi(1)a^2 + \frac{1}{4}a^4)\pi_S$ with $a = \delta/(2\sigma_S)$.*

The leading term of (3) is the value of a group: per unit, the log of the ratio of its peak density π_*/σ_* to the noise density π_0/R , less the entropy $\frac{1}{2} \log(2\pi e) \approx 1.42$ of a unit normal error. With $R/\sigma_* = 40$, $\pi_* = 0.1$, $\pi_0 = 0.2$ and $\xi \approx 0$, the value is 1.58 nats per unit of the group, that is 0.158n in all. With these values the loss term is $\nu \log 2$: for a fit whose

noise mass off G is about 0.1, the weight $\pi_0 = 0.2$ less the group’s 0.1, it is about $0.069n$ and the gain at least $0.09n$ against a margin $\frac{d+1}{2} \log n$ of 8.6 for $d = 2$, $n = 300$. The bound $\nu \leq 1 - \pi_*$ alone gives a loss of up to $0.62n$, which exceeds the value, so the inequality is informative only for such a fit. Part (ii), an idealisation with two parallel lines at intercepts δ apart, says that removing one of two lines through one group is nearly free, so a candidate that arises by the two operations is accepted for all large n when the right side of (3) exceeds $\varphi(1)a^2\pi_S$ and its scales are admissible. The result is for fixed parameters, which the step reads off the data, and it does not cover the reweighting refit of the candidate. The peak test, analysed in Supplement S2.3, fails a band of outliers that fills its window and passes one of half-width below about 3.4 scales.

4. Simulation study

4.1. Design

Data. Each data set has $n = 300$ units, a covariate uniform on $[-3, 3]$ and standard normal errors. The eight designs are: D1, two crossing lines with equal weights and noise; D2, three parallel lines three noise scales apart ($m = 12$); D3, as D1 with weights 0.7 and 0.3 ($m = 10$); D4, as D1 with noise scales 0.5 and 1.5; D5, as D1 with high-leverage outliers; D6, as D1 with uniform outliers; D7, as D1 with two further covariates of slope 1 ($m = 12$); D8, two parallel lines six noise scales apart; $m = 8$ unless stated. In D1, D3, D4 and D6 the two lines cross at the origin with slopes ± 1.5 , so the groups overlap near the crossing and even the Bayes classifier with known parameters labels about 9% of the units wrongly; an accuracy of about 0.91 is the ceiling there, and essentially one in D8. Each unit is replaced by an outlier independently with probability ε , so the realised fraction of outliers varies around ε . Outliers in the response are shifted up or down by a uniform amount between 15 and 30 times the average noise scale; high-leverage outliers (D5) have their covariate redrawn uniformly on $[6, 9]$, outside the design range, and their response uniformly on $[-3, 3]$; uniform outliers (D6) are spread over the bounding box of the clean data enlarged by 20%, away from the true lines.

Competitors. The main rival is TCLUST-REG (García-Escudero et al., 2010), from the FSDA toolbox (Riani et al., 2012), run unmodified under GNU Octave with restriction factor 12, 1000 random starts, 50 refinement steps and no second-level trimming in the covariates, at the two conservative levels $\alpha = 0.25$ and 0.40, followed by reweighting and then by the recovery step. The reweighting attached to TCLUST-REG is our own rule, the map of Section 3 applied twice with $c = 2.5$ and a median absolute deviation per group, which carries the idea of Dotto et al. (2018) to regression; it is not the incremental scheme of that paper, which is examined separately (Supplement, Section S6). “TCLUST-REG, reweighted” denotes this construction throughout. The Supplement (Tables S4 to S11, S37 and S38) holds the rows of a shorter search (300 starts, 10 refinement steps, $\alpha \in \{0.05, 0.10, 0.15, 0.25, 0.40\}$), which understated TCLUST-REG with three groups, and those of the trimmed cluster-weighted model (García-Escudero et al., 2017) and the trimmed likelihood estimator (Neykov et al., 2007) at the same five levels. CTLE (Chang et al., 2020) and the bisquare mixture of regressions (Bai et al., 2012) were the competitors without a level from published software; they come from the R package RobMixReg (Cao et al., 2026), as do the two trimmed methods.

Four further competitors without a level were added after the first part had been analysed: the mixture with Laplace errors of Song et al. (2014); the mixture of contaminated normal regressions (Punzo & McNicholas, 2017; Mazza & Punzo, 2020); a mixture with a uniform noise component in the manner of Banfield & Raftery (1993), the last two our implementations; and sequential RANSAC (Fischler & Bolles, 1981), which extracts the K lines one at a time and is given the true noise scale, so it is an oracle benchmark, not a competitor that can be run on data. Least squares with 50 random starts is a reference without protection (software and settings: Supplement, Section S1). The monitoring approach of Torti et al. (2019) leaves the choice of the level to the analyst; it enters the comparison only through two automatic rules for reading a level off the monitoring path (Section 4.4).

Criteria. Every method is given the true number of groups K and returns K coefficient vectors, from which every clean unit receives the label of its nearest fitted line. The *accuracy* is the fraction of clean units whose label agrees with the true one, maximised over relabellings; trimmed and flagged units are treated alike, and a failed fit counts as accuracy zero (failures were rare except for CTLE; Supplement, Section S5). This labelling ignores the estimated scales and weights, so it is not the Bayes rule of any method where the groups differ in scale, as in D4, and it does not penalise trimming or flagging clean units as such. The coefficient error, the flagged fraction and the computing time are therefore reported as well (Section 4.4), and so are the adjusted Rand index on all units, with the outliers as a class of their own, and a misclassification rate that counts a flagged clean unit as an error. By these two criteria the main pipelines rank as they do by accuracy: up to $\varepsilon = 0.20$ the mean adjusted Rand index is 0.719 for ESF and 0.725 and 0.715 for TCLUST-REG at levels 0.25 and 0.40 with reweighting and the recovery step, beyond it 0.660 against 0.519 and 0.714 (Supplement, Table S45). Labelling the clean units by TCLUST-REG’s own assignment instead of the nearest line changed its mean accuracy in D2, D4 and with four groups by -0.03 to $+0.02$ (Supplement, Table S49), so the criterion does not drive the comparison.

Analysis plans. The study was run in parts, and before each part a written plan fixed the comparison to be made and its thresholds (Plans 1 to 5; Supplement, Table S47, with the digests of the plan files in Table S34); the third part is descriptive. Four things reported below were decided after all the plans: the main rival pipeline, TCLUST-REG with the long search, reweighting and the recovery step; the three mixtures and sequential RANSAC; the best-of-five rule of Section 2.5; and a guard in the scoring scale of ESF for groups fitted exactly (Section 5.3), which changed nothing on the 40 simulated data sets on which we checked it.

4.2. Up to 20% of outliers

Table 1 shows, for each design and method, the lowest mean accuracy over $\varepsilon \in \{0, 0.05, 0.10, 0.20\}$; if a method works only at the right trimming level, or only with equal groups, at least one entry in its row is low. Supplement Table S40 gives the TCLUST-REG rows cell by cell, with the paired differences from ESF.

The rows of TCLUST-REG show what a generous level costs and what the recovery step returns. At level 0.40 with reweighting, TCLUST-REG trims away the smaller group whenever the groups are unequal: its worst accuracy is 0.66 in D3, where the smaller group

Table 1: Worst mean accuracy on clean units (the lowest of the cell means) with up to 20% of outliers ($\varepsilon \in \{0, 0.05, 0.10, 0.20\}$), by design. D1 to D8: first part (50 data sets per cell). $K = 4$, small groups (80/20 and 85/15, including the fresh data of Plan 4), clustered outliers and other designs (sample size, moderate outliers, dependent covariates, t_3 , skewed and heteroscedastic errors): third part and Plans 4 and 5. Column entries are minima over different sets of ε : D1 to D8 over the four values above, the other columns over the values of $\varepsilon \leq 0.20$ run in the third part and Plans 4 and 5. TCLUST-REG was run with 1000 random starts and 50 refinement steps (the rows with the level 0.30 and with equal weights were run after the others); “reweighted” is our two-pass rule ($c = 2.5$, median absolute deviation per group). Sequential RANSAC is given the true noise scale and is an oracle benchmark, not a competitor that can be run on data. CTLE and the mixtures were run with the starts of Supplement Section S1. In D5, TCLUST-REG at level 0.25 with 5% second-level trimming in the covariates and reweighting (300 starts) reaches 0.86 (Supplement, Table S37). Values below 0.85 in italics.

Method	D1	D2	D3	D4	D6	D7	D8	D5	$K = 4$	Small	Clust.	Other
ESF (with the recovery step)	0.91	0.89	0.90	0.90	0.89	0.91	1.00	0.85	0.95	<i>0.83</i>	<i>0.78</i>	0.90
ESF with $m = 12$ in every design	0.91	0.90	0.89	0.90	0.89	0.91	1.00	0.86	0.88	<i>0.83</i>	<i>0.75</i>	0.90
ESF alone	0.91	0.89	0.89	0.90	0.90	0.91	1.00	0.85	0.95	<i>0.60</i>	<i>0.70</i>	0.88
TCLUST-REG, $\alpha = 0.25$, reweighted, then the recovery step	0.91	0.89	0.91	0.90	0.91	0.91	1.00	<i>0.83</i>	0.91	0.90	0.90	0.91
TCLUST-REG, $\alpha = 0.40$, reweighted, then the recovery step	0.91	<i>0.83</i>	0.91	0.90	0.91	0.91	1.00	0.87	<i>0.81</i>	0.85	0.91	0.89
TCLUST-REG, $\alpha = 0.30$, reweighted, then the recovery step	0.91	0.87	0.91	0.90	0.91	0.91	1.00	0.87	0.88	0.89	0.91	0.90
TCLUST-REG, $\alpha = 0.25$, equal weights, reweighted, then the recovery step	0.91	0.90	0.91	0.90	0.91	0.91	1.00	0.86	0.96	0.89	0.90	0.90
TCLUST-REG, $\alpha = 0.25$, reweighted	0.91	0.89	0.89	0.90	0.91	0.91	1.00	<i>0.83</i>	0.91	<i>0.60</i>	0.90	0.88
TCLUST-REG, $\alpha = 0.40$, reweighted	0.91	<i>0.83</i>	<i>0.66</i>	0.90	0.91	0.91	1.00	0.87	<i>0.81</i>	<i>0.56</i>	0.91	<i>0.60</i>
CTLE	0.91	<i>0.59</i>	0.87	0.85	0.87	0.89	<i>0.72</i>	<i>0.73</i>	<i>0.55</i>	<i>0.64</i>	0.86	<i>0.81</i>
Bisquare mixture	0.89	<i>0.56</i>	0.88	0.90	0.90	0.90	<i>0.80</i>	<i>0.55</i>	<i>0.47</i>	<i>0.81</i>	<i>0.57</i>	0.85
Laplace mixture	0.91	<i>0.61</i>	0.91	0.91	0.91	0.91	0.96	<i>0.58</i>	<i>0.60</i>	0.88	<i>0.63</i>	0.90
Contaminated normal mixture	0.91	<i>0.52</i>	0.91	0.91	0.91	0.91	1.00	<i>0.83</i>	<i>0.65</i>	0.91	<i>0.58</i>	0.89
Noise-component mixture	0.91	<i>0.75</i>	0.91	0.90	0.91	0.90	<i>0.81</i>	<i>0.53</i>	<i>0.65</i>	<i>0.64</i>	<i>0.61</i>	0.88
Sequential RANSAC (oracle: true noise scale given)	0.91	<i>0.69</i>	0.91	0.90	0.91	0.91	0.97	<i>0.54</i>	<i>0.71</i>	0.91	<i>0.84</i>	0.90
Least squares (50 starts)	<i>0.53</i>	<i>0.37</i>	<i>0.70</i>	<i>0.52</i>	<i>0.58</i>	<i>0.53</i>	<i>0.53</i>	<i>0.53</i>	<i>0.49</i>	<i>0.80</i>	<i>0.61</i>	<i>0.51</i>

holds 30% of the units, and 0.56 with groups of 15% and 20%. The recovery step returns these groups (0.91 and 0.85) but does not repair three and four groups, where the level discards part of several groups at once (0.83 and 0.81 with or without the step). At level 0.25 with reweighting and the recovery step, TCLUST-REG is at 0.89 or more in every design except D5 (0.83); without the step it loses the small groups at $\varepsilon = 0$ (0.60). TCLUST-REG was run without second-level trimming in the covariates, so in D5 the high-leverage points can be trimmed only through their residuals, hence the 0.83; the adaptive form of that trimming did not help either (Supplement, Section S6.1). ESF is at 0.85 or more in every design except clustered outliers at $\varepsilon = 0.20$ (0.78) and groups of 80% and 20% with three covariates at $\varepsilon = 0.20$ (0.83). With four groups it reaches 0.95, against 0.91 for the level 0.25 and 0.81 for the level 0.40 with estimated weights, and is matched only by TCLUST-REG at level 0.25 with equal weights (0.96). CTLE and the four mixtures all fall below 0.85 with three or four groups and with high-leverage points. With three groups CTLE and the Laplace and bisquare mixtures are below least squares with 50 starts even without outliers (0.78, 0.66 and 0.72 against 0.90), so their default search is part of the failure; a start from least squares raised the two mixtures we implemented to 0.89 with three groups and 0.95 with four without outliers, but not with outliers, and 100 starts did not help (Supplement, Tables S21 and S27). Sequential RANSAC, although given the true noise scale, falls to 0.69 with three parallel groups and 0.71 with four.

Table 2 sets the three pipelines against the choice an analyst has to make. With up to 20% of outliers, the better of the two levels, chosen after the run, never falls below 0.85, while the worse one does so in five cells. ESF, which chooses nothing, is within 0.03 of the better of the levels 0.25 and 0.40 in 83 of the 89 cells, above it by more than 0.03 in

Table 2: The choice of a trimming level: mean over cells of the mean accuracy on clean units, the number of cells with a mean below 0.85, and the worst cell, for all 123 cells of the study (Plans 1 to 5 and the third part; the $\varepsilon = 0.30$ cells of the first part are excluded, as the second part replaces them). “Better” and “worse” of the two levels are chosen in each cell after the run, between TCLUST-REG at 0.25 and at 0.40, each with reweighting and the recovery step. The last four rows were run after the others: the levels 0.30 and 0.35, and the levels 0.25 and 0.40 with equal weights (option `equalweights` of FSDA), all with the long search.

Method	$\varepsilon \leq 0.20$ (89 cells)			$\varepsilon \geq 0.25$ (34 cells)		
	Mean	Below 0.85	Worst	Mean	Below 0.85	Worst
ESF	0.914	2	0.78	0.820	13	0.53
ESF, $m = 12$ in every design	0.912	2	0.75	0.813	13	0.53
TCLUST-REG, $\alpha = 0.25$, rw, recovery	0.917	1	0.83	0.693	26	0.51
TCLUST-REG, $\alpha = 0.40$, rw, recovery	0.909	4	0.81	0.863	8	0.53
Better of the two levels, chosen afterwards	0.918	0	0.87	0.865	7	0.53
Worse of the two levels	0.908	5	0.81	0.691	27	0.51
TCLUST-REG, $\alpha = 0.25$, rw, no recovery	0.890	11	0.60	0.695	26	0.51
TCLUST-REG, $\alpha = 0.40$, rw, no recovery	0.827	31	0.56	0.869	7	0.54
TCLUST-REG, $\alpha = 0.30$, rw, recovery	0.915	0	0.87	0.794	17	0.53
TCLUST-REG, $\alpha = 0.35$, rw, recovery	0.912	4	0.83	0.851	10	0.53
TCLUST-REG, $\alpha = 0.25$, equal weights, rw, recovery	0.919	0	0.86	0.769	24	0.53
TCLUST-REG, $\alpha = 0.40$, equal weights, rw, recovery	0.907	4	0.79	0.856	10	0.54

two (four groups at $\varepsilon = 0.10$ and 0.20) and below it by more than 0.03 in four: clustered outliers and the three small-group designs of Plans 3 and 4 at $\varepsilon = 0.20$, with gaps of 0.03 to 0.125. Without the recovery step, the level 0.40 falls below 0.85 in 31 cells, 27 of them with unequal or small groups. These counts describe design points chosen to probe weaknesses, not frequencies in any population of data sets. With 200 data sets in each of the eleven cells that decide the comparison (150 added; Supplement, Table S50), the standard errors of the paired differences fall to about 0.01 and the conclusions stand: with four groups ESF is ahead of TCLUST-REG at levels 0.25 and 0.30 by 0.03 to 0.06 and on a par with the level 0.25 with equal weights, and with a small group or clustered outliers at $\varepsilon = 0.20$ it is behind the level 0.25 by 0.03 to 0.09.

The levels in between and equal weights trade one range of contamination for the other (Table 2, last four rows). With reweighting and the recovery step, the level 0.30 falls below 0.85 in no cell up to 20% of outliers and in 17 of the 34 beyond, the level 0.35 in four and ten. Equal weights make the level 0.25 the most accurate pipeline up to 20% (mean 0.919, no cell below 0.85; ESF is within 0.03 of it in 85 of the 89 cells and below it by more in four) but leave 24 cells below 0.85 beyond. They change the summaries of the level 0.40 little, although cell by cell they gain 0.01 to 0.05 with three groups and 0.04 to 0.05 with four and lose up to 0.20 with unequal or small groups. The two pipelines that do best up to 20% appear as rows of Table 1: the level 0.30 has no entry below 0.85 and reaches 0.88 with four groups, where the level 0.25 with equal weights reaches 0.96 and ESF 0.95. Over all 123 cells the level 0.40 with reweighting and the recovery step has the highest mean (0.897), then the level 0.35 (0.895), the level 0.40 with equal weights (0.893) and ESF (0.888). By mean accuracy and by the number of cells within 0.03 of ESF up to 20%, the level 0.35 is the closest to a level that serves both ranges: up to 20% of outliers it is within 0.03 of ESF in 83 of the 89 cells (mean 0.912 against 0.914), ahead in two and behind in four, three of them

Table 3: Worst mean accuracy on clean units with 25% to 35% of outliers. D1 to D8: second part, new data (100 data sets per cell); the other columns: third part at $\varepsilon = 0.30$ only (50 data sets per cell), so column entries are minima over different sets of ε . Rows and conventions as in Table 1. Values below 0.85 in italics.

Method	D1	D2	D3	D4	D6	D7	D8	D5	$K = 4$	Small	Clust.	Other
ESF (with the recovery step)	0.91	<i>0.55</i>	<i>0.84</i>	0.90	<i>0.58</i>	0.88	1.00	<i>0.53</i>	<i>0.72</i>	<i>0.75</i>	<i>0.55</i>	0.87
ESF with $m = 12$ in every design	0.91	<i>0.56</i>	<i>0.83</i>	0.89	<i>0.55</i>	0.87	0.98	<i>0.53</i>	<i>0.77</i>	<i>0.66</i>	<i>0.56</i>	0.87
ESF alone	0.91	<i>0.56</i>	<i>0.84</i>	0.90	<i>0.59</i>	0.88	1.00	<i>0.53</i>	<i>0.73</i>	<i>0.73</i>	<i>0.54</i>	0.87
TCLUST-REG, $\alpha = 0.25$, reweighted, then the recovery step	<i>0.53</i>	<i>0.51</i>	<i>0.70</i>	<i>0.53</i>	<i>0.69</i>	<i>0.53</i>	<i>0.53</i>	<i>0.53</i>	<i>0.74</i>	<i>0.82</i>	<i>0.59</i>	<i>0.52</i>
TCLUST-REG, $\alpha = 0.40$, reweighted, then the recovery step	0.91	<i>0.76</i>	0.87	0.90	<i>0.72</i>	0.90	1.00	<i>0.53</i>	<i>0.76</i>	<i>0.82</i>	<i>0.70</i>	0.91
TCLUST-REG, $\alpha = 0.30$, reweighted, then the recovery step	<i>0.60</i>	<i>0.61</i>	<i>0.74</i>	<i>0.63</i>	<i>0.73</i>	<i>0.59</i>	<i>0.60</i>	<i>0.53</i>	<i>0.77</i>	<i>0.86</i>	<i>0.62</i>	<i>0.77</i>
TCLUST-REG, $\alpha = 0.25$, equal weights, reweighted, then the recovery step	<i>0.59</i>	<i>0.66</i>	<i>0.71</i>	<i>0.57</i>	<i>0.74</i>	<i>0.57</i>	<i>0.64</i>	<i>0.53</i>	<i>0.78</i>	<i>0.82</i>	<i>0.61</i>	<i>0.75</i>
TCLUST-REG, $\alpha = 0.25$, reweighted	<i>0.53</i>	<i>0.51</i>	<i>0.70</i>	<i>0.53</i>	<i>0.78</i>	<i>0.53</i>	<i>0.53</i>	<i>0.53</i>	<i>0.74</i>	<i>0.82</i>	<i>0.55</i>	<i>0.52</i>
TCLUST-REG, $\alpha = 0.40$, reweighted	0.91	<i>0.76</i>	0.87	0.90	0.90	0.90	1.00	<i>0.54</i>	<i>0.76</i>	<i>0.81</i>	<i>0.68</i>	0.91
CTLE	<i>0.65</i>	<i>0.40</i>	<i>0.65</i>	<i>0.47</i>	<i>0.67</i>	<i>0.55</i>	<i>0.66</i>	<i>0.51</i>	<i>0.47</i>	<i>0.62</i>	<i>0.67</i>	<i>0.69</i>
Bisquare mixture	<i>0.73</i>	<i>0.42</i>	<i>0.82</i>	<i>0.76</i>	<i>0.83</i>	<i>0.63</i>	<i>0.75</i>	<i>0.53</i>	<i>0.46</i>	<i>0.80</i>	<i>0.57</i>	<i>0.68</i>
Laplace mixture	0.91	<i>0.48</i>	0.90	0.91	0.90	0.90	0.94	<i>0.53</i>	<i>0.53</i>	0.86	<i>0.61</i>	0.90
Contaminated normal mixture	0.90	<i>0.37</i>	0.91	0.90	<i>0.67</i>	<i>0.65</i>	0.85	<i>0.61</i>	<i>0.48</i>	0.88	<i>0.53</i>	<i>0.71</i>
Noise-component mixture	<i>0.55</i>	<i>0.40</i>	<i>0.72</i>	<i>0.55</i>	0.87	<i>0.54</i>	<i>0.53</i>	<i>0.52</i>	<i>0.46</i>	<i>0.83</i>	<i>0.57</i>	<i>0.56</i>
Sequential RANSAC (oracle: true noise scale given)	0.91	<i>0.64</i>	0.91	0.91	<i>0.73</i>	0.91	0.98	<i>0.53</i>	<i>0.71</i>	0.89	<i>0.63</i>	0.90
Least squares (50 starts)	<i>0.53</i>	<i>0.37</i>	<i>0.70</i>	<i>0.53</i>	<i>0.53</i>	<i>0.53</i>	<i>0.53</i>	<i>0.53</i>	<i>0.47</i>	<i>0.81</i>	<i>0.62</i>	<i>0.51</i>

with four groups, where it loses 0.11 to 0.13; beyond 25% it is ahead of ESF in 12 of the 34 cells and behind in four. Table 2 is descriptive: these levels were run after the analysis plans.

The criterion of Plan 1, in the seven designs without high-leverage points at $\varepsilon \leq 0.20$, 28 cells, was met in every cell by ESF alone, the largest shortfall against the best competitor being 0.029 (standard error 0.014), and again by ESF with the recovery step under Plan 3 (Supplement, Table S1 and Section S6). Its second criterion, that every fixed-level competitor trail ESF alone by at least 0.05 in some cell, was not met: TCLUST-REG at level 0.25 with reweighting was never more than 0.033 below it, so this competitor was tested again, on new data, at higher contamination.

4.3. Beyond 25% of outliers

Plan 2 asked whether TCLUST-REG at level 0.25 with reweighting fails once the contamination exceeds its level while ESF alone does not. The answer was yes, by a wide margin and by construction: at $\varepsilon = 0.30$ and 0.35 this competitor fell to accuracies between 0.53 and 0.62 in D1, D4, D7 and D8, with the short search and with the long one (Table 3), where ESF alone stayed above 0.87.

The level 0.40 is the more robust choice from 25% of outliers. With reweighting and the recovery step it is ahead of ESF by more than 0.03 in 13 of the 34 cells and never behind by as much (Table 2; Figure 2). ESF breaks down in four situations. With uniform background noise (D6) it falls to 0.83 at $\varepsilon = 0.25$ and 0.58 at 0.35, while TCLUST-REG at level 0.40 with reweighting and the Laplace mixture stay at 0.90, and the recovery step, applied after TCLUST-REG, lowers this to 0.72 (Section 4.6). With three groups (D2) it falls to 0.76 at $\varepsilon = 0.30$ and 0.55 at 0.35, against 0.86 and 0.76 for the level 0.40, and with four groups at 0.30 they reach 0.72 and 0.76. With high-leverage outliers (D5) it falls to 0.64 at $\varepsilon = 0.25$ and about 0.53 beyond, where no method is reliable, the best of Supplement Table S11 being the trimmed cluster-weighted model at level 0.40 (0.83 at $\varepsilon = 0.25$, 0.70 at 0.35). With a small group or clustered outliers at 30% it reaches 0.75 and 0.55, against 0.82 to 0.87 and 0.70 for the level 0.40. In D1, D4, D7 and D8 it holds up to 35% of outliers. The Laplace mixture is the strongest method without a level at high contamination, at 0.90 or more up

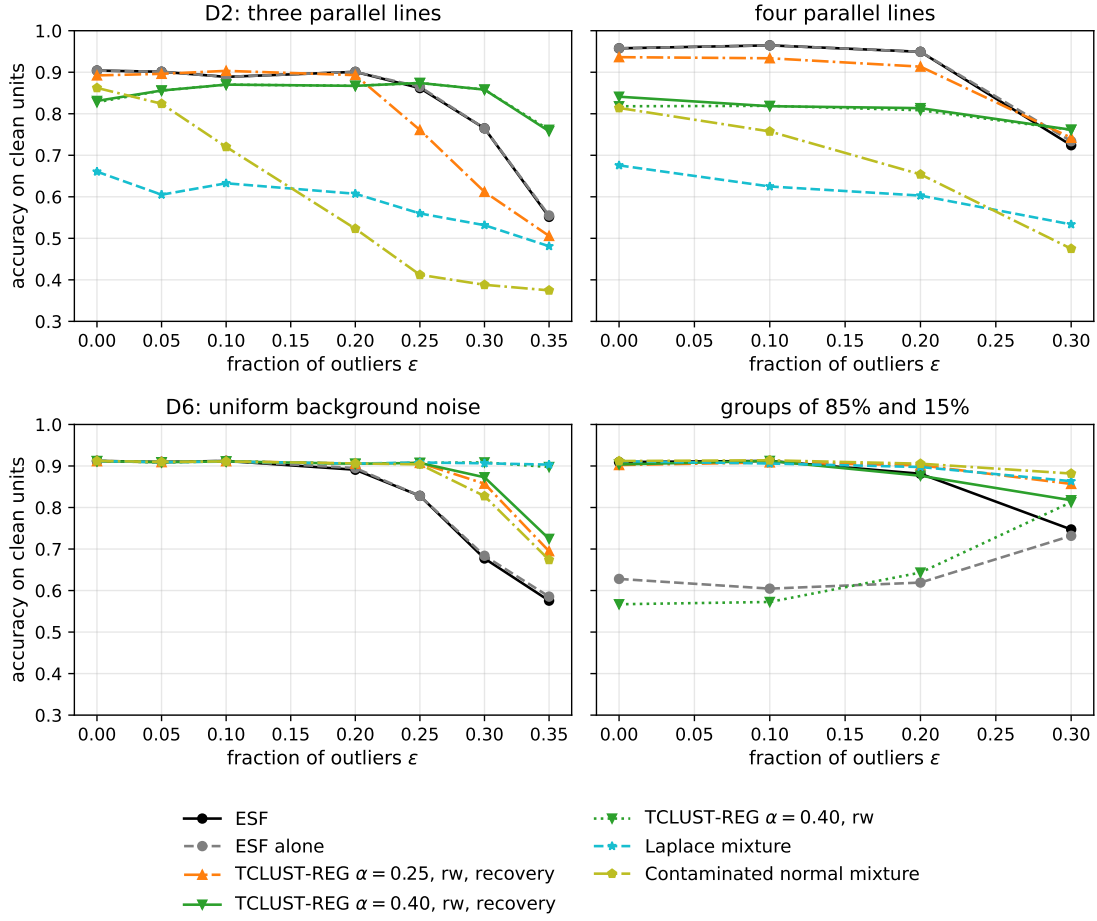


Figure 2: Mean accuracy on clean units against the fraction of outliers in four designs. D2 and D6: first part ($\epsilon \leq 0.20$, 50 data sets each) and second part ($\epsilon \geq 0.25$, 100 data sets each, new data); four groups and groups of 85% and 15%: third part (50 data sets each), which was run at $\epsilon \leq 0.30$ only. TCLUST-REG with 1000 starts; rw: reweighted by our rule; recovery: followed by the recovery step.

to $\varepsilon = 0.35$ in D1, D3, D4, D6, D7 and D8, but it fails with three and four groups already without outliers (0.66 and 0.68) and with clustered outliers.

4.4. Covariate screen, other measures, further checks and cost

The covariate screen. It is what makes ESF robust to high-leverage points. Without it (Supplement, Tables S37 and S38) the first step breaks down in D5 at $\varepsilon = 0.20$, as least squares does; outside D5 the two versions differ by at most 0.03 up to $\varepsilon = 0.20$ in the first two parts, and from $\varepsilon = 0.25$ on the screen gains up to 0.13 and costs at most 0.02. The coefficient error (Supplement, Table S19) tells the same story as the accuracy: up to $\varepsilon = 0.20$ the median error of ESF is at most 0.24 in every design D1 to D8, and it exceeds 1 only where its accuracy has broken down.

The flagged fraction. Where ESF works, its mean flagged fraction $\hat{\alpha}$ (Supplement, Table S31) lies within about three percentage points of ε , and the flagged set contains at least 89% of the outliers and about 3% of the clean units or fewer (Supplement, Table S16); where ESF breaks down it falls well below ε , because the outliers are absorbed into a fitted line, which is the warning sign. After TCLUST-REG at either conservative level, reweighting and the recovery step, the flagged fraction is as close to ε as that of ESF (mean absolute deviation over the 89 cells with $\varepsilon \leq 0.20$: 0.016 and 0.017 at the two levels, against 0.018).

Further checks. Supplement Section S6 and Table S39 report TCLUST-REG with equal scales, with adaptive second-level trimming in the covariates and with a level chosen from its monitoring path, the mixture of t regressions (Yao et al., 2014), robust linear grouping (García-Escudero et al., 2009), the incremental reweighting of Dotto et al. (2018), two changes to our own constants, and the mixtures with more starts (Tables S21 and S27). None gave a TCLUST-REG without a level or did better than the pipelines above: the monitoring rules chose either a level below the contamination or, nearly always, 0.40, and the incremental reweighting fell below 0.85 in 22 cells with $\varepsilon \leq 0.20$ after the recovery step, against four for the single step.

Cost. On one core ESF took a median of 0.08 seconds per data set and about two seconds with three covariates, most of it in the covariate screen; the times of the competitors are in Supplement Table S32.

4.5. Sample size, groups, covariates and outliers

The third, descriptive part of the study (2,300 data sets, 50 per cell) varied what the first two held fixed: the sample size ($n = 100$ and $n = 1000$); the number of groups (four parallel lines four noise scales apart, $n = 400$, $m = 16$); dependent covariates; moderate outliers shifted by 4 to 8 noise scales, and outliers in a tight cloud; groups of 80% and 20% ($m = 16$) and of 85% and 15% ($m = 20$); and t_3 errors. Plan 5 added skewed errors, groups of 70% and 30% with three covariates ($m = 16$), and an error scale growing fourfold over the design range (Supplement, Table S25 and Section S6). The sample size, dependent covariates and moderate outliers change little: ESF stays within 0.014 of the best competitor chosen afterwards, except that with moderate outliers in D3 at $\varepsilon = 0.30$ it trails TCLUST-REG at level 0.25 by 0.04. Small groups are what the recovery step is for: ESF alone reaches only 0.82 without outliers and 0.73 at $\varepsilon = 0.20$ with groups of 80% and 20%, and between 0.60

and 0.73 with groups of 85% and 15%, because its best-scoring fit puts two nearly parallel lines on the large group and flags most of the small one (Section 2.3). With the recovery step it reaches 0.88 to 0.91 up to $\varepsilon = 0.20$, within 0.035 of the best competitor, while at $\varepsilon = 0.30$ the first step flags more than a quarter of the units, the step rarely applies, and the accuracy stays at 0.75. Heavy-tailed, skewed and heteroscedastic errors leave the accuracy within 0.006 of the best competitor in every cell, but the flagged fraction then counts the tails or the spread of the errors as well as the outliers: it exceeds the contamination by 6.5% of the units with t_3 errors and no outliers and by 1 to 6 percentage points in Plan 5. Clustered outliers are the hard case: at $\varepsilon = 0.20$ ESF alone fits the cloud with one of its two lines and falls to 0.70; the recovery step repairs 11 of the 27 fits that fail in this way and raises the mean to 0.78, against 0.90 for TCLUST-REG at a level above the contamination, and at $\varepsilon = 0.30$ no method of Table 3 exceeds 0.72.

4.6. *The recovery step: confirmation, sensitivity and repeated runs*

The recovery step was developed on data generated with other seeds and frozen before Plan 3, which applied it to the stored fits of ESF alone on the 6,400 data sets the study then held. All three criteria were met: in the six small-group cells with $\varepsilon \leq 0.20$ the step raised the mean accuracy by 0.07 to 0.31 (at least 0.05 was required), in no cell with $\varepsilon \leq 0.20$ did it lower it by more than 0.004 (0.01 was allowed), and the comparison of Plan 1 was passed (Supplement, Table S36). On the 600 new data sets of Plan 4, with two small-group designs not tried before (Supplement, Table S41), the step raised the accuracy by 0.09 to 0.28 in all 12 cells, but with 20% of outliers in the two new designs ESF was 0.08 and 0.09 below TCLUST-REG at level 0.25 with reweighting and the recovery step (0.91 and 0.97); in the other ten cells it was within 0.024 of it.

Sensitivity and repeated runs. Moving each constant of the recovery step one step down and one step up (Supplement, Tables S23 and S44) changed no cell mean by more than about 0.03, except the limit on the flagged fraction (Section 2.5). The noise range R , 1.1 times the range of the response, is set by the most extreme units, and a larger range makes the uniform component less likely, so that a band of outliers passes more easily as a group. With R doubled, as if the most extreme unit lay twice as far out, the mean accuracy of ESF up to 20% of outliers moved from 0.914 to 0.912, but clustered outliers at $\varepsilon = 0.20$ lost 0.09; with R five times larger the losses reached 0.14 with uniform noise at $\varepsilon = 0.20$, and after TCLUST-REG at level 0.40 up to 0.29 beyond 25% (Supplement, Table S48). Two ranges tied to the fitted scales, 20 median scales and the range of the unflagged units plus 10 median scales, remove this dependence but make the uniform component more likely everywhere and lose small groups: the mean accuracy of ESF up to 20% fell to 0.891 and 0.903, with 12 and 9 cells below 0.85 against two. The range of the paper is the best of the three on these designs, at the price of a dependence on the most extreme units. ESF depends on its seed: in five runs with different seeds on each data set, the accuracies differed by more than 0.05 on 3.5% of the data sets with up to 20% of outliers and on 13% of all data sets, most of them where ESF breaks down (Supplement, Section S6); this variation is of the size of the thresholds of the plans, so the paired comparisons resolve differences of about 0.03 and no finer.

The recovery step after a trimming method. After TCLUS-REG at level 0.40 with reweighting, the recovery step repairs the failures of the level with unequal groups up to $\varepsilon = 0.20$ (Section 4.2; Supplement, Table S40) but not with three or four groups, and it harms it materially once in the study (elsewhere by at most 0.016): with uniform noise at 30% and 35% it lowers the accuracy from 0.91 to 0.87 and from 0.90 to 0.72, on 47 of the 100 data sets at 35% by more than 0.05, because a dense and narrow band of uniform noise passes the peak test (Section 3). Moving each constant of the step did not prevent it (Supplement, Section S5). After the trimmed cluster-weighted model with reweighting (Supplement, Table S46) the step did the same: up to $\varepsilon = 0.20$ it reduced the cells below 0.85 from 39 to 12 at level 0.25 and from 66 to 20 at level 0.40, and it lowered a cell mean by more than 0.02 only with uniform noise at 30% and 35% (by up to 0.06) and once with two parallel lines at 30% (0.025).

5. Three data sets

5.1. The fishery data

The fishery data (Riani et al., 2008), distributed with FSDA, record 677 monthly import flows of one fishery product into the European Union from a third country: the quantity and the value of each flow. They are the standard example for TCLUS-REG and the forward search (Perrotta et al., 2009; Cerioli & Perrotta, 2014): flows priced differently form lines through the origin, and a number of flows fit none of them. We work on the logarithmic scale, where $\text{value} = \text{price} \times \text{quantity}$ becomes a line of slope one whose intercept is the log price, the spread of the errors no longer grows with the quantity, and the concentration of small flows near the origin, which Cerioli & Perrotta (2014) handle by thinning, disappears. Twenty-six flows with zero quantity or value have no logarithm and are left out, which leaves 651. The published analyses (Perrotta et al., 2009; Cerioli & Perrotta, 2014; Torti et al., 2019) keep them and work on the raw scale, where the heteroscedasticity and the dense region near the origin are part of the problem; the analysis below therefore answers a different question and is not a comparison with their results. We fit $K = 2$ groups with the constants of Section 2.5 and $m = 8$.

Supplement Table S42 and Figure S1 show the results. ESF alone finds two lines of slope close to one, with intercepts 2.57 and 2.85, that is prices of about 13 and 17, and flags 11% of the flows, about 80% of them priced below both lines. The recovery step finds that these cheap flows lie close to a line of their own: for 19 of 20 seeds it replaces the line at 2.57 by a line with intercept 2.16, a price of about 9, keeps one line through the two dearer price levels (intercept 2.78), and flags 3% of the flows.

TCLUS-REG gives the same two answers, but which one depends on the trimming level: at levels up to 0.08 it treats the cheap flows as a group, with intercepts between 2.2 and 2.3, as ESF does; from level 0.20 on it separates the two dearer price levels, as ESF alone does, and its estimates then vary from one random start to another (Supplement, Section S5). With the long search, reweighting and the recovery step, level 0.25 returns the cheap group and the dearer line (intercepts 2.15 to 2.16 and 2.78, 3% of the flows flagged) from all ten starts, and level 0.40 from three of ten; without the recovery step both levels separate the two dearer price levels.

The data suggest three price levels. With $K = 3$ and $m = 16$, ESF returns lines with intercepts 1.90 to 1.91, 2.50 to 2.54 and 2.82 to 2.83, slopes between 0.94 and 1.00, and 3% of the flows flagged, for 12 of 20 seeds; the other seeds give other fits. The best-of-five rule of Section 2.5 returned the cheap group and the dearer line with $K = 2$, and the three price levels with $K = 3$, every time over twenty disjoint sets of seeds (Supplement, Section S5). The Bayesian information criterion of the same model prefers $K = 2$ (Section 6); whether the cheap flows are a price level or anomalous declarations is a question for the subject-matter expert.

5.2. *The tone perception data*

The tone perception data (De Veaux, 1989), distributed with the R package `mixtools` (Benaglia et al., 2009), record 150 trials in which a musician tuned a tone to the octave above a fundamental with stretched overtones; the tuned ratio lies either near 2 or near the stretch ratio, so the data form a nearly flat line and a line of slope one. Bai et al. (2012) added ten identical points at $(0, 4)$, far from both lines and outside the range of the covariate. We fit $K = 2$ with the constants of Section 2.5 and $m = 8$, to the original data and with the ten points added (Supplement, Table S35). On the original data every method finds the two lines, except the trimmed cluster-weighted model at level 0.25, which returns other fits from two of its ten starts. With the ten points added, 6% of the enlarged sample, one of the two lines is lost from every start by TCLUST-REG at levels 0.02 and 0.05, by the trimmed cluster-weighted model at level 0.05 and by the Laplace mixture; the trimmed cluster-weighted model at levels 0.10 and 0.25 loses it from three and six of its ten starts, and the noise-component mixture for 7 of 20 seeds. ESF returns the same two lines, to within 0.01, for every seed, as do CTLE, the contaminated normal mixture and TCLUST-REG from level 0.10 on, with either search; the recovery step found no group among the flagged trials. The flagged fraction of ESF, 0.25 to 0.27 on the original data, is not a count of gross outliers: a tight core of trials sets the robust scale of the rising line, so 18 to 20 trials of that line are flagged by their residuals, and the covariate screen flags 14 more (Supplement, Section S5); TCLUST-REG with reweighting flags 19%. The fitted lines are not affected.

5.3. *Taxi fares from an airport*

The third example has a known answer. The New York City Taxi and Limousine Commission publishes every yellow-taxi trip (NYC Taxi and Limousine Commission, 2024); we took the 157,264 trips of March 2024 that started at John F. Kennedy airport and carry a rate code. Their fare, before tolls, surcharges and tips, follows one of two tariffs: a flat fare of 70 dollars to Manhattan (rate code 2 with a valid record, 53% of the trips), or the meter, which charges by distance and, in slow traffic, by time (rate code 1, 38%); the remaining 9.6% use other tariffs or are invalid records (Supplement, Section S5) and are treated as contamination. We regressed the fare on the distance and the duration, $d = 3$, with $K = 2$ and $m = 10$; nothing was removed, and the rate code gives the true group of every clean trip. Twenty random samples of 2,000 trips were fitted by every method of Section 4 except sequential RANSAC, which needs the noise scale. The flat-fare group lies exactly on its line, so more than half of the units have residual zero and the scale of the scoring step, a median of squared residuals, is zero; the guard mentioned in Section 4.1 takes the median over the units that a replicate does not fit exactly.

Supplement Table S43 gives the results. ESF found both tariffs in all twenty samples: the flat fare exactly, and a metered line with median intercept 2.63 dollars, 3.04 dollars per mile and 0.33 dollars per minute, against 2.40, 3.06 and 0.33 for the fit to all metered trips. Its accuracy of 0.978 is the ceiling that the best methods share: TCLUST-REG and the trimmed cluster-weighted model at levels 0.10 to 0.40, the bisquare, contaminated normal and noise-component mixtures all reach 0.975 to 0.979, as does TCLUST-REG with the long search, reweighting and the recovery step at both levels, with smaller coefficient errors (0.10 and 0.14 against 0.23); the recovery step changed none of its fits. TLE at levels 0.10 and 0.15 and CTLE stopped with an error on most samples, presumably because the flat-fare group has no variance; the example shows that the methods hold on a group without noise, not that one is better than another. ESF took a median of 31 seconds per sample, most of it in the minimum covariance determinant of the covariate screen, for which we used a general-purpose implementation (Supplement, Table S43). Its flagged fraction, 0.23, is more than twice the contamination: the metered fares depend on the time spent in slow traffic, not on the duration, and their residuals have heavy tails, so 16% of the clean trips are flagged, more than half of them by the covariate screen, which reads the long, skewed trips as outlying covariates; 89% of the contaminated trips are flagged. TCLUST-REG with reweighting, which has no covariate screen, flags 0.14. The best-of-five rule of Section 2.5 failed here once, because with a group that has no noise the likelihood rewards any line through tied points.

6. Discussion

The recovery step asks whether the units a robust fit has discarded hide a group, by the likelihood of Gaussian groups plus uniform noise, and it can follow any method that flags units. After TCLUST-REG at a generous level with reweighting it returned the groups of 15% to 30% that the level had trimmed away, and it reduced the cells with a mean accuracy below 0.85, up to 20% of outliers, from 31 of 89 to four; the four that remain have three or four groups, where the level discards part of several groups at once, and the step left them at 0.83 and 0.81. After the trimmed cluster-weighted model it did the same. No fixed level of TCLUST-REG between 0.25 and 0.40, with or without equal weights, was the best choice on both sides of 20% of outliers; the level 0.35 came closest, and was within 0.03 of ESF in 83 of the 89 cells up to 20% but lost 0.11 to 0.13 with four groups. ESF, the flagging procedure without a trimming level, is the option when that knowledge is missing and the contamination is probably below a quarter. On the three data sets ESF reached, without a level, the answers that TCLUST-REG gives at the right level.

The step settles whether a poorly fitted set of units is contamination or a group, by the likelihood of the model that the flagging rule already assumes, one candidate at a time and at the given K ; where the number of groups is in doubt, a scan over the number of groups (Section 2.4) is the better tool. After a generous level the step accepted narrow bands of uniform noise as a group at 30% and 35% of outliers, so it should not follow such a level when a large fraction of uniform noise is possible.

The flagged fraction $\hat{\alpha}$ is the quantity a user of a level-free method reads first, and it is a contamination estimate only under near-Gaussian errors. Proposition 2 says what it estimates then, the contamination plus about 1.2% of clean units, and on the simulated data

with Gaussian errors it lay within about three percentage points of ε wherever ESF worked. On the tone data ESF flagged 0.25 to 0.27 of a benchmark that is treated as clean, and on the taxi data 0.23 against a contamination of 9.6%, because the errors are far from Gaussian in both (Section 5); the fitted lines were right in both cases. A flagged fraction well above any plausible contamination therefore says that the errors are not Gaussian, and one well below it says that the fit has broken down (Section 2.5).

Set ESF against TCLUST-REG at level 0.30 with reweighting and the recovery step. Up to 20% of outliers the two have the same mean accuracy over the 89 cells, 0.914 against 0.915, and the fixed level has no cell below 0.85 where ESF has two, at 0.78 and 0.83 (Table 2). ESF gives three things. It needs no level, and the level 0.30 is a choice that pays only because the contamination in those cells is below it: beyond 25% the same level falls below 0.85 in 17 of 34 cells, ESF in 13. With four groups it is more accurate, 0.95 against 0.88 (Table 1), because the trimming level discards part of several groups at once and the estimated group weights make this worse; the level 0.25 with equal weights removes that loss (0.96) but not the loss beyond 25%. It costs three things: a subsample size m , which encodes a bound on the smallest group proportion or, at $m = 12$, gives up a little accuracy (Section 2.5); the cap of $n/3$ on the flagged set, which stops the recovery step where the fit is in doubt and so limits the recovery of small groups to about 20% of outliers; and a dependence on the seed that the fit of TCLUST-REG with 1000 starts does not show, which the best-of-five rule reduces at five times the cost.

The choice among the methods of the study is as follows. If the contamination is known to be below a fifth, TCLUST-REG at level 0.25 with equal weights, reweighting and the recovery step was the most accurate pipeline, with ESF within 0.03 of it in 85 of 89 cells. If it may exceed a quarter, use a level of 0.35 or 0.40 with reweighting, followed by the recovery step unless the outliers may be spread uniformly. If nothing is known about it, ESF with the recovery step needs no level and is accurate up to about a fifth, and a flagged fraction well below the plausible contamination warns that it has broken down. With two groups and outliers in the response only, the Laplace mixture was about as accurate as ESF up to 20% of outliers and more robust beyond, but it fails with three or four groups, with high-leverage points and with clustered outliers.

Five limits should be kept in mind. ESF breaks down when outliers are so frequent that clean subsamples are rare in the first stages and a contaminated fit is taken for the reference; in our designs this happened from 25% of outliers with uniform background noise or high-leverage points and from 30% with three groups, while with two groups of equal size, or with three covariates, it held up to 35%. Small groups are recovered only up to about 20% of outliers, and not on every data set at 20% (Section 4.6). Outliers concentrated in a tight cloud were fitted as part of a group from 20% on; the recovery step repaired about two fifths of these fits. If the error spread grows with the covariate, units with large covariate values are flagged: with a spread growing fourfold over the design range the flagged fraction exceeded the contamination by up to 6 points (Plan 5), and in raw trade data the spread grows much faster, so a transformation that stabilises it, such as the logarithm of Section 5, should precede the fit. The candidate search of the recovery step, a line through d random units among 500 draws, is itself unstable when the group to be recovered is a small part of the flagged set: on the fishery data, after TCLUST-REG at level 0.40 with reweighting, the step found the cheap group from three of ten starts (Section 5.1), so the step can miss a

group that it would accept. Its noise range is set by the most extreme units, so that outliers far out make bands of noise easier to accept as groups (Section 4.6). A result for the whole procedure would need to control which units are flagged, and we do not have one (Section 3).

The number of groups K was taken as known. The Bayesian information criterion of the Gaussian-lines-plus-noise model over several values of K gives a way to choose it: in a descriptive check (Supplement, Table S18) it chose the true K for at least 90% of the data sets with two or four groups and up to 20% of outliers, but for only 50% to 80% with the three overlapping groups of D2, and it counted a tight cloud of outliers as a group; on the fishery data it prefers $K = 2$ to $K = 1$ and $K = 3$.

Declarations

Funding The author received no specific funding for this work.

Competing interests The author declares no competing interests.

Ethics approval Not applicable.

Data availability The fishery data are distributed with the FSDA toolbox (Riani et al., 2012), and the tone perception data with the R package `mixtools` (Benaglia et al., 2009). The taxi data are public trip records of the New York City Taxi and Limousine Commission (NYC Taxi and Limousine Commission, 2024); the extract used and the raw simulation output are in the repository named under Code availability and in its Zenodo archive.

Code availability The code, the scripts that produce every table and figure, and the analysis plans with their SHA-256 digests are at <https://github.com/sorujov/eesp> (“eesp” is the name of an earlier version of the method), archived at Zenodo (concept DOI <https://doi.org/10.5281/zenodo.22807276>, which resolves to the latest version).

Use of large language models During the preparation of this work the author used Claude (Anthropic), a large language model, to assist with writing and checking code, running and managing the simulations, and drafting and revising the text. The author reviewed and edited all such content and takes full responsibility for the article; the model is not an author.

References

- Bagirov, A. M., Ugon, J., & Mirzayeva, H. (2013). Nonsmooth nonconvex optimization approach to clusterwise linear regression problems. *European Journal of Operational Research*, 229(1), 132–142.
- Bai, X., Yao, W., & Boyer, J. E. (2012). Robust fitting of mixture regression models. *Computational Statistics & Data Analysis*, 56(7), 2347–2359.
- Banfield, J. D., & Raftery, A. E. (1993). Model-based Gaussian and non-Gaussian clustering. *Biometrics*, 49(3), 803–821.
- Benaglia, T., Chauveau, D., Hunter, D. R., & Young, D. S. (2009). `mixtools`: an R package for analyzing finite mixture models. *Journal of Statistical Software*, 32(6), 1–29.

- Bertsimas, D., & Shioda, R. (2007). Classification and regression via integer optimization. *Operations Research*, 55(2), 252–271.
- Brusco, M. J. (2006). A repetitive branch-and-bound procedure for minimum within-cluster sums of squares partitioning. *Psychometrika*, 71(2), 347–363.
- Cao, S., Chang, W., & Zhang, C. (2026). *RobMixReg: Robust Mixture Regression*. R package version 1.1.3, published on CRAN 2 June 2026. doi:10.32614/CRAN.package.RobMixReg.
- Carbonneau, R. A., Caporossi, G., & Hansen, P. (2011). Globally optimal clusterwise regression by mixed logical-quadratic programming. *European Journal of Operational Research*, 212(1), 213–222.
- Carbonneau, R. A., Caporossi, G., & Hansen, P. (2012). Extensions to the repetitive branch and bound algorithm for globally optimal clusterwise regression. *Computers & Operations Research*, 39(11), 2748–2762.
- Carbonneau, R. A., Caporossi, G., & Hansen, P. (2014). Globally optimal clusterwise regression by column generation enhanced with heuristics, sequencing and ending subset optimization. *Journal of Classification*, 31(2), 219–241.
- Ceroli, A., García-Escudero, L. A., Mayo-Iscar, A., & Riani, M. (2018). Finding the number of normal groups in model-based clustering via constrained likelihoods. *Journal of Computational and Graphical Statistics*, 27(2), 404–416.
- Ceroli, A., & Perrotta, D. (2014). Robust clustering around regression lines with high density regions. *Advances in Data Analysis and Classification*, 8(1), 5–26.
- Chang, W., Zhou, X., Zang, Y., Zhang, C., & Cao, S. (2020). Component-wise adaptive trimming for robust mixture regression. arXiv preprint arXiv:2005.11599.
- DeSarbo, W. S., & Cron, W. L. (1988). A maximum likelihood methodology for clusterwise linear regression. *Journal of Classification*, 5(2), 249–282.
- De Veaux, R. D. (1989). Mixtures of linear regressions. *Computational Statistics & Data Analysis*, 8(3), 227–245.
- Dotto, F., Farcomeni, A., García-Escudero, L. A., & Mayo-Iscar, A. (2018). A reweighting approach to robust clustering. *Statistics and Computing*, 28(2), 477–493.
- Fischler, M. A., & Bolles, R. C. (1981). Random sample consensus: a paradigm for model fitting with applications to image analysis and automated cartography. *Communications of the ACM*, 24(6), 381–395.
- Fois, A., Insolia, L., Consolini, L., Laurini, F., Locatelli, M., & Riani, M. (2026). Clusterwise linear regression using a probabilistic branch and bound algorithm under Gaussianity. *Computers & Operations Research*, 189, 107375.
- García-Escudero, L. A., Gordaliza, A., San Martín, R., Van Aelst, S., & Zamar, R. (2009). Robust linear clustering. *Journal of the Royal Statistical Society Series B*, 71(1), 301–318.

- García-Escudero, L. A., Gordaliza, A., Mayo-Iscar, A., & San Martín, R. (2010). Robust clusterwise linear regression through trimming. *Computational Statistics & Data Analysis*, 54(12), 3057–3069.
- García-Escudero, L. A., Gordaliza, A., Matrán, C., & Mayo-Iscar, A. (2011). Exploring the number of groups in robust model-based clustering. *Statistics and Computing*, 21(4), 585–599.
- García-Escudero, L. A., Gordaliza, A., Greselin, F., Ingrassia, S., & Mayo-Iscar, A. (2017). Robust estimation of mixtures of regressions with random covariates, via trimming and constraints. *Statistics and Computing*, 27(2), 377–402.
- Gervini, D., & Yohai, V. J. (2002). A class of robust and fully efficient regression estimators. *The Annals of Statistics*, 30(2), 583–616.
- Gurobi Optimization, LLC (2026). *Gurobi Optimizer Reference Manual*, version 13.0. <https://www.gurobi.com>.
- Joki, K., Bagirov, A. M., Karmitsa, N., Mäkelä, M. M., & Taheri, S. (2020). Clusterwise support vector linear regression. *European Journal of Operational Research*, 287(1), 19–35.
- Kuhn, H. W. (1955). The Hungarian method for the assignment problem. *Naval Research Logistics Quarterly*, 2(1–2), 83–97.
- Magri, L., & Fusiello, A. (2014). T-linkage: a continuous relaxation of J-linkage for multi-model fitting. In *2014 IEEE Conference on Computer Vision and Pattern Recognition* (pp. 3954–3961).
- Magri, L., & Fusiello, A. (2015). Robust multiple model fitting with preference analysis and low-rank approximation. In *Proceedings of the British Machine Vision Conference 2015* (pp. 20.1–20.12). BMVA Press.
- Maher, S., Miltenberger, M., Pedroso, J. P., Rehfeldt, D., Schwarz, R., & Serrano, F. (2016). PySCIPOpt: mathematical programming in Python with the SCIP Optimization Suite. In *Mathematical Software – ICMS 2016*, Lecture Notes in Computer Science 9725 (pp. 301–307). Cham: Springer.
- Mazza, A., & Punzo, A. (2020). Mixtures of multivariate contaminated normal regression models. *Statistical Papers*, 61(2), 787–822.
- Megiddo, N., & Tamir, A. (1982). On the complexity of locating linear facilities in the plane. *Operations Research Letters*, 1(5), 194–197.
- Neykov, N., Filzmoser, P., Dimova, R., & Neytchev, P. (2007). Robust fitting of mixtures using the trimmed likelihood estimator. *Computational Statistics & Data Analysis*, 52(1), 299–308.
- NYC Taxi and Limousine Commission (2024). *TLC Trip Record Data*: yellow taxi trip records, March 2024. <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page> (accessed 27 September 2026).
- Park, Y. W., Jiang, Y., Klabjan, D., & Williams, L. (2017). Algorithms for generalized clusterwise linear regression. *INFORMS Journal on Computing*, 29(2), 301–317.
- Perrotta, D., Riani, M., & Torti, F. (2009). New robust dynamic plots for regression mixture detection. *Advances in Data Analysis and Classification*, 3(3), 263–279.

- Pollard, D. (1981). Strong consistency of k -means clustering. *The Annals of Statistics*, 9(1), 135–140.
- Punzo, A., & McNicholas, P. D. (2017). Robust clustering in regression analysis via the contaminated Gaussian cluster-weighted model. *Journal of Classification*, 34, 249–293.
- Quandt, R. E. (1972). A new approach to estimating switching regressions. *Journal of the American Statistical Association*, 67(338), 306–310.
- Riani, M., Cerioli, A., Atkinson, A. C., Perrotta, D., & Torti, F. (2008). Fitting mixtures of regression lines with the forward search. In F. Fogelman-Soulié et al. (Eds.), *Mining Massive Data Sets for Security* (pp. 271–286). Amsterdam: IOS Press.
- Riani, M., Perrotta, D., & Torti, F. (2012). FSDA: A MATLAB toolbox for robust analysis and interactive data exploration. *Chemometrics and Intelligent Laboratory Systems*, 116, 17–32.
- Rousseeuw, P. J. (1984). Least median of squares regression. *Journal of the American Statistical Association*, 79(388), 871–880.
- Rousseeuw, P. J., & Leroy, A. M. (1987). *Robust Regression and Outlier Detection*. New York: Wiley.
- Rousseeuw, P. J., & Van Driessen, K. (1999). A fast algorithm for the minimum covariance determinant estimator. *Technometrics*, 41(3), 212–223.
- Rousseeuw, P. J., & Van Driessen, K. (2006). Computing LTS regression for large data sets. *Data Mining and Knowledge Discovery*, 12(1), 29–45.
- Song, W., Yao, W., & Xing, Y. (2014). Robust mixture regression model fitting by Laplace distribution. *Computational Statistics & Data Analysis*, 71, 128–137.
- Späth, H. (1979). Algorithm 39: Clusterwise linear regression. *Computing*, 22(4), 367–373.
- Toldo, R., & Fusiello, A. (2008). Robust multiple structures estimation with J-linkage. In *Computer Vision – ECCV 2008*, Lecture Notes in Computer Science 5302 (pp. 537–547). Berlin: Springer.
- Torti, F., Perrotta, D., Riani, M., & Cerioli, A. (2019). Assessing trimming methodologies for clustering linear regression data. *Advances in Data Analysis and Classification*, 13(1), 227–257.
- Torti, F., Riani, M., & Morelli, G. (2021). Semiautomatic robust regression clustering of international trade data. *Statistical Methods & Applications*, 30(3), 863–894.
- Yao, W., Wei, Y., & Yu, C. (2014). Robust mixture regression using the t -distribution. *Computational Statistics & Data Analysis*, 71, 116–127.
- Yu, C., Yao, W., & Chen, K. (2017). A new method for robust mixture regression. *Canadian Journal of Statistics*, 45(1), 77–94.